Agent-orchestrated engineering: batch first, verify natively

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One engineer + an AI agent + open analysis: GPU exploration with native verification on NVIDIA GB10.

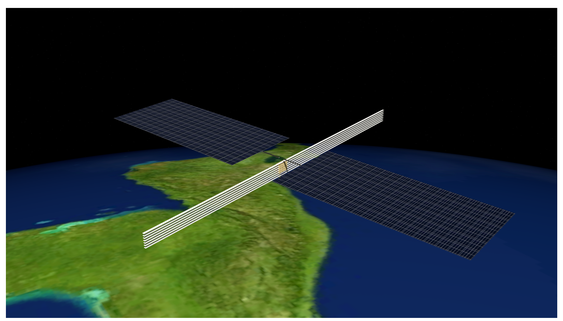
8.49 million designs/s; 33.1× warm sweep speedup vs one GB10 Cortex-X925 CPU core.

≈11-day documented build | 1.81M designs screened | 1.05M uncertainty scenarios | 2 release-audit gates

Human requirements / design decisions → AI agent: inference closed-product black box → Open analysis scripts Apache-2.0 → GPU: Warp 1–3; PhysicsNeMo 4; Newton 5 4: train → infer; CPU: CalculiX FEA; speed 6 → Native numerical checks → Human approval gates A + B

Verified mass-and-thermal trade point

30 × 250 kW = 7.5 MW installed peak
175 kW nominal (sunlit); annual 164.4 kW/sc
92.674 t incl. 2-t dispenser; 7.326 t reserve
140 panels; 861.75 m² gross array
GPU/HBM 366.16 / 361.84 K nominal
Peak inlet 391.91 K [BOL; ε = 0.92]



Earth: NASA Blue Marble; rendered in Isaac Sim (RTX path tracing); illustrative

Designed / screened

861.75 m² solar array.
140-panel Al 6061-T6.
7 round 8/7-mm tubes.
MPG40; 31.25-kg pack.
Eclipse compute pause.
30-envelope stow; FEA.

Not designed / OPEN

MMOD: 694 / 5 yr.
EOL: −16.53 kW/sc.
Peak GPU: 404.0 K.
Flight loads / stow.
Flex / force tests.
Radiation; ADCS / links.

So what? A verified trade point

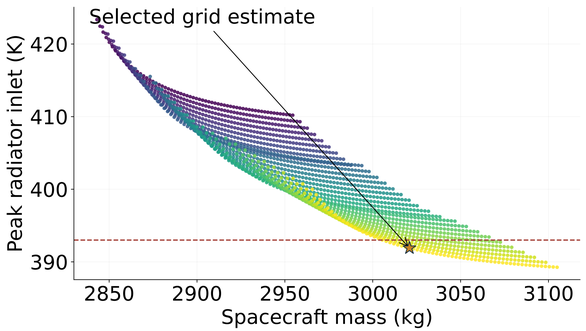
Screened 1.81M designs, quantified uncertainty, verified natively; exposed OPEN limits in ≈11 days.

Open, reproducible analogue: Dassault / Synopsys showed agentic engineering at GTC Berlin 2026.

23-mm seams 1.495%: 319.7 → 324.55 W/m² active. Fill / annual OPEN; freeze FAIL waived, mass charged.

1 Warp | discrete sizing

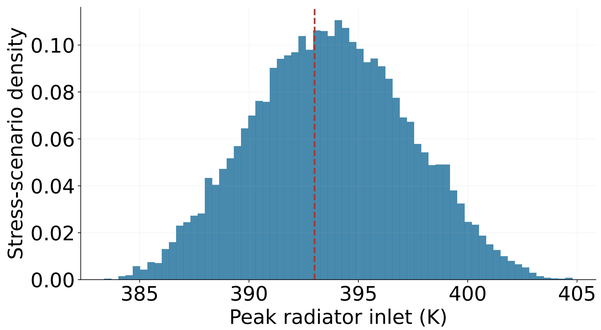
1,806,336 designs



Minimum full mass in verified stow subset. Star: surrogate estimate; native check: 3022.48 kg/sc, 140 panels.

2 Warp | stress uncertainty

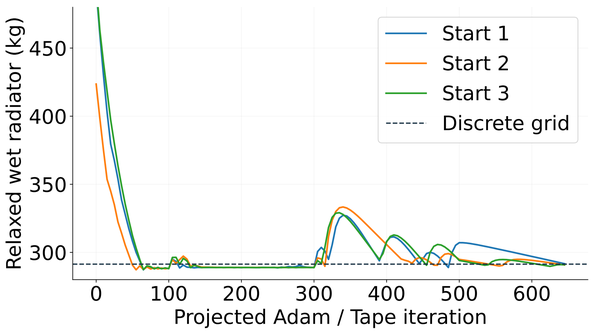
42.08% scenario pass fraction



1,048,576 bounded engineering scenarios + rank sensitivity, not measured reliability. Coating spec: ε = 0.92 (BOL).

3 warp.Tape | gradients

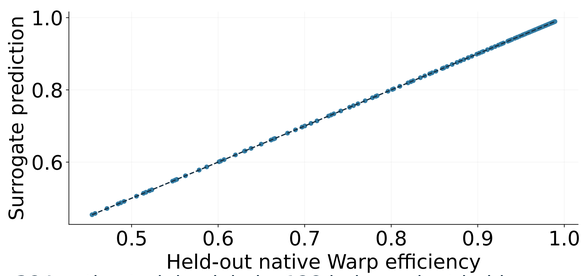
0.82% relaxed mass gap



Finite-difference gradient check PASS. Continuous relaxation cross-checks the grid; not adopted hardware or a global optimum.

4 PhysicsNeMo | held out

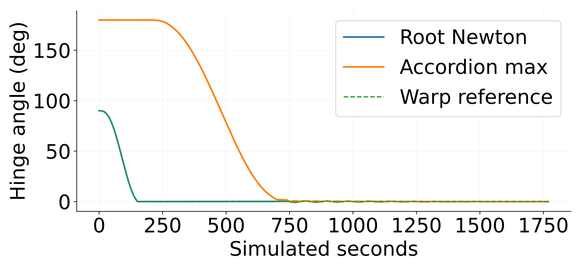
Max |Δη| = 9.3e-05



384 native training labels; 128 independent held-out labels. Selected fins re-solved natively.

5 Newton | rigid deployment

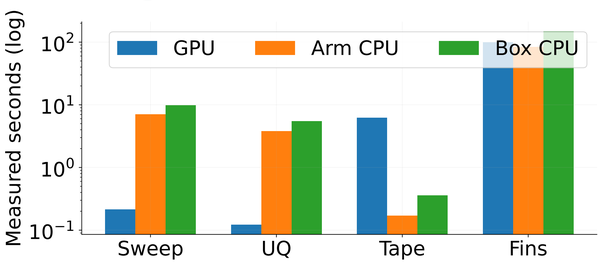
31 / 31 representative locks



Sampled clearance PASS; angle parity 0.0061 deg. Force-step convergence and flexible hardware OPEN. Both solvers on CUDA.

6 Honest speed | same fp64 work

Batching wins; serial work can lose



Same fp64 kernel; CPU: 1 core / 1 thread, not all 20 Arm cores. Warm runs exclude setup/JIT/training; serial GPU work can lose.

Hardware: GB10 GPU; CPU: 1 Cortex-X925 core / 1 thread; box: x86 VM. Warp 1.18.0 / 1.19.0; Newton 1.7.0; PhysicsNeMo 2.2.0; CalculiX 2.21.
https://github.com/neilndave/orbital-compute-reverse-engineered | References: SpaceX; Warp; Newton; PhysicsNeMo; Dow; NASA NTRS 20030068423.
Apache-2.0 analysis scripts; proprietary AI agent and Isaac Sim renderer. AI assistance: Grok Bot; author responsible. Not flight qualification.
Canonical SHA256: f02f82201c45c9b1c98b1052f8adc795f47c441e642059bd450b288b5de88846